MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 2 Solution

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1. Suppose X_i are nonempty sets of numbers. Let $X = \bigcup_i X_i$ and $\lambda_i = \sup X_i$. Prove that $\sup X = \sup\{\lambda_i\}$ if it exits.

   Solution.
   Let $K = \sup_i \lambda_i$, we show that it is an upper bound of X. For any $x \in X$, $x \in X_i$ for some i and $x \le \lambda_i \le K$. Now if $K \neq \sup X$, $K - \sup X = 2\epsilon > 0$. There is λ_i such that $K - \lambda_i < \epsilon$ and there is x_i such that $\lambda_i - x_i < \epsilon$. We have

   $$K - x_i \le K - \lambda_i + \lambda_i - x_i < 2\epsilon \implies \sup X < x_i,$$

   a contradiction. ■

2. Let the set $A = \left\{\frac{1}{3} \pm \frac{n}{3n+1} \mid n \in \mathbb{N}\right\}$. Prove that $\inf A = 0$ and $\sup A = \frac{2}{3}$.

   Solution.
   $\frac{n}{3n+1} < \frac{1}{3}$ for all n, so 0 is a lower bound. For any $\epsilon > 0$,

   $$\epsilon > \frac{1}{3} - \frac{n}{3n+1} = \frac{1}{3(3n+1)}$$
   $$9n + 3 > \frac{1}{\epsilon}$$

   So, when $n > \frac{1}{\epsilon} \implies 9n + 3 > \frac{1}{\epsilon}$, we have $\epsilon > \frac{1}{3} - \frac{n}{3n+1}$. 0 is the infimum.
   $\frac{1}{3} + \frac{n}{3n+1} = \frac{6n+1}{3(3n+1)} < \frac{2}{3}$, $\frac{2}{3}$ is an upper bound. For any $\epsilon > 0$,

   $$\epsilon > \frac{2}{3} - \frac{6n+1}{3(3n+1)} = \frac{1}{3(3n+1)}$$

   Again, take $n > \frac{1}{\epsilon}$. $\frac{2}{3}$ is the supremum. You can also consider symmetry to finish the proof. ■

3. Let $P(\mathbb{N})$ be the set that contains all subsets of $\mathbb{N}$. Is it countable?

   Solution.
   Suppose there is a bijection $f : \mathbb{N} \to P(\mathbb{N})$. i.e. $f(n) = U_n$ for any $n \in \mathbb{N}$ and U_n is a subset of $\mathbb{N}$. Now define $V = \{n \mid n \notin U_n\}$, f is a bijection implies there is $k \in \mathbb{N}$ s.t. $f(k) = U_k = V$. But, $k \in U_k \implies k \notin V = U_k$, $k \notin U_k \implies k \in V = U_k$, a contradiction. Hence, $P(\mathbb{N})$ is not countaable.

   ■

4. Prove the following are equivalent to the definition of $\lim_{n\to\infty} x_n = L$.

   (a) For any $c > \epsilon > 0$, where c is some fixed number, there is $N \in \mathbb{N}$, such that $|x_n - L| < \epsilon$ for all $n > N$.

   (b) For any $\epsilon > 0$, there is $N \in \mathbb{N}$, such that $|x_n - L| \leq \epsilon$ for all $n > N$.

   (c) For any $\epsilon > 0$, there is $N \in \mathbb{N}$, such that $|x_n - L| < \epsilon$ for all $n \geq N$.

   (d) For any $\epsilon > 0$, there is $N \in \mathbb{N}$, such that $|x_n - L| < 2\epsilon$ for all $n > N$.

   (e) For any $\epsilon > 0$, there is $N \in \mathbb{N}$, such that $|x_n - L| < \epsilon^2$ for all $n > N$.

   (f) For any natural number K, there is $N \in \mathbb{N}$, such that $|x_n - L| < \frac{1}{K}$ for all $n > N$.

   Solution.
   We just show (d). Assume the definition of limit. For any $\epsilon > 0$, consider $\epsilon_0 = 2\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $|x_n - L| < \epsilon_0 = 2\epsilon$ for all $n > N$.
   Now assume (d). For any $\epsilon > 0$, consider $\epsilon_1 = \epsilon/2$, there is $N \in \mathbb{N}$ s.t. $|x_n - L| < 2\epsilon_1 = \epsilon$ for all $n > N$. They are equivalent. ■

5. (a) Prove that if a_n is a sequence converging to L, then the sequence $|a_n|$ of absolute values converges to $|L|$.

   (b) Suppose now that the sequence $|a_n|$ of absolute values converges to $|L|$. Is it necessarily that a_n converges to L?

   Solution.

   (a) For any $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $||a_n| - |L|| \leq |a_n - L| < \epsilon$ for all $n > N$.

   (b) It is true when $L = 0$ only, take $a_n = (-L)^n$ when $L \neq 0$

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6. (a) Let a_n and b_n be sequences. Prove that if $\lim_{n\to\infty} a_n = 0$ and b_n is bounded, then $\lim_{n\to\infty} a_n b_n = 0$.

   (b) Suppose now that $\lim_{n\to\infty} a_n = L \neq 0$ and b_n is bounded. What can we conclude about the convergence of the sequence $a_n b_n$?

   Solution.

   (a) Since b_n is bounded, there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. When $a_n < \frac{\epsilon}{M}$,

   $$|a_n b_n| < |a_n| M < M \cdot \frac{\epsilon}{M} = \epsilon$$

   (b) If $a_n = b_n = 1$ for all n, $a_n b_n$ converges. If $b_n = (-1)^n$, $a_n b_n$ diverges.

■

7. Use the definition of convergence of a sequence to prove that

$$\lim_{n\to\infty} \frac{n^2+n+1}{3n^2+2} = \frac{1}{3}$$

Solution.

$$\left|\frac{n^2+n+1}{3n^2+2} - \frac{1}{3}\right| = \frac{3n+1}{3(3n^2+2)} < \frac{3n+3n}{3(3n^2+0)} < \frac{2}{3n} < \frac{1}{n} < \frac{1}{N} < \epsilon$$

For any $\epsilon > 0$, there is $N \in \mathbb{N}$ s.t. $N > \frac{1}{\epsilon}$, $n > N \implies \left|\frac{n^2+n+1}{3n^2+2} - \frac{1}{3}\right| < \epsilon$. ■

— *End* —